

LINDESNES FYR, Norway

WMO: 014360

Lat: 57.983N

Lon: 7.048E

Elev: 13

StdP: 101.17

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.0	-5.0	-12.5	1.3	-5.6	-10.1	1.6	-3.3	21.6	5.1	20.0	1.5	8.7	50	0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	21.4	17.0	20.2	16.4	19.1	15.9	17.9	20.5	17.1	19.4	16.4	18.5	5.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	19.3	16.0	11.4	18.6	15.3	10.9	17.9	50.7	20.7	48.2	19.4	46.1	18.7	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	17.4	15.8	DB	-6.8	23.0	2.5	1.8	-8.7	24.2	-10.1	25.2	-11.6	26.2	-13.4	27.5	(4)
(5)			WB	-7.9	18.7	2.4	1.3	-9.7	19.6	-11.0	20.3	-12.4	21.1	-14.1	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	2.4	1.7	3.4	6.3	10.5	13.2	16.0	16.0	13.7	10.0	6.4	4.2	(6)	
	DBStd	5.74	3.41	3.29	2.79	2.23	3.00	2.38	2.28	1.55	1.86	2.36	3.37	3.74	(7)	
	HDD10.0	1133	234	233	206	113	27	1	0	0	1	28	111	179	(8)	
	HDD18.3	3528	493	466	464	361	243	154	81	76	138	257	358	437	(9)	
	CDD10.0	660	0	0	0	2	43	98	186	185	112	29	3	0	(10)	
	CDD18.3	12	0	0	0	0	1	1	8	2	0	0	0	0	(11)	
	CDH23.3	5	0	0	0	0	0	0	4	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.0	9.4	8.3	7.9	6.8	6.8	7.3	7.1	7.4	8.2	8.6	8.9	8.9	(14)	
(15) Precipitation	PrecAvg	1300	141	108	79	75	67	78	88	118	128	147	136	134	(15)	
	PrecMax	1659	294	207	179	133	138	164	209	209	229	250	285	212	(16)	
	PrecMin	779	19	14	14	31	12	7	16	22	50	47	40	30	(17)	
	PrecStd	212	57	57	39	27	33	44	45	45	50	60	54	51	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	7.8	9.7	13.3	20.8	21.4	23.7	21.9	19.1	15.5	12.7	10.5	(19)	
		MCWB	7.8	6.6	6.5	8.8	16.1	15.2	18.7	17.1	16.3	13.4	11.5	9.3	(20)	
	2%	DB	8.1	7.0	8.0	12.0	18.8	19.9	22.2	20.2	17.8	14.5	12.1	9.7	(21)	
		MCWB	7.0	5.9	5.2	8.8	14.3	14.9	17.7	16.8	15.5	12.9	11.0	8.6	(22)	
	5%	DB	7.5	6.5	7.1	10.9	16.9	18.4	20.9	19.3	17.1	13.8	11.3	9.2	(23)	
		MCWB	6.4	5.5	5.5	8.0	12.9	14.6	17.2	16.3	15.0	12.2	10.0	8.2	(24)	
	10%	DB	6.9	5.9	6.4	9.6	14.9	16.8	19.6	18.4	16.4	13.2	10.6	8.6	(25)	
		MCWB	5.9	5.0	5.2	7.2	11.7	13.8	16.6	15.9	14.4	11.7	9.2	7.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	7.0	7.3	10.3	16.7	16.9	19.7	18.4	16.9	14.3	12.1	9.6	(27)	
		MCDB	8.7	7.4	8.7	12.1	20.4	19.6	23.0	20.2	18.0	15.1	12.5	10.1	(28)	
	2%	WB	7.2	6.2	6.4	9.2	14.7	15.7	18.5	17.5	16.1	13.5	11.2	8.9	(29)	
		MCDB	7.8	6.7	7.2	11.0	18.3	18.6	21.3	19.4	17.3	14.3	12.0	9.5	(30)	
	5%	WB	6.6	5.7	5.9	8.4	13.2	14.9	17.6	16.8	15.5	12.7	10.2	8.3	(31)	
		MCDB	7.3	6.2	6.6	10.3	16.0	17.6	20.4	18.6	16.8	13.5	11.1	9.0	(32)	
	10%	WB	6.0	5.1	5.4	7.5	12.0	14.1	16.8	16.2	14.8	12.0	9.5	7.6	(33)	
		MCDB	6.9	5.8	6.2	9.3	14.5	16.4	19.2	18.0	16.3	13.0	10.4	8.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	2.9	3.1	3.4	3.6	3.6	3.4	3.4	3.2	3.1	3.0	3.0	(35)	
		MCDBR	3.1	2.7	3.4	4.4	5.1	4.7	4.7	4.1	3.6	3.0	2.7	2.6	(36)	
	5% WB	MCWBR	3.6	2.8	2.6	3.0	3.7	3.5	3.6	2.9	3.1	3.0	3.2	3.1	(37)	
		MCWBR	3.1	2.6	2.9	3.8	4.9	4.2	4.2	3.4	3.1	2.9	2.8	2.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.338	0.354	0.389	0.376	0.365	0.380	0.385	0.364	0.345	0.319	0.294	(40)	
		taud	2.206	2.290	2.338	2.291	2.404	2.460	2.437	2.420	2.464	2.404	2.271	2.172	(41)	
	Ebn at Noon		512	688	791	818	859	874	850	817	778	677	508	415	(42)	
		Edh at Noon	51	73	91	111	106	102	102	97	81	65	47	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.34	0.94	2.30	3.88	5.25	5.87	5.32	4.07	2.64	1.26	0.47	0.24	(44)	
	RadStd		0.07	0.19	0.24	0.43	0.61	0.53	0.56	0.33	0.27	0.19	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	-134	-183	+62	N/A

Nomenclature: See separate page